MICHAEL JEFFERSON

FULL-STACK PRODUCT ENGINEER & APPLIED AI BUILDER

Vacaville, CA (US Remote & SF Bay Area Hybrid) | Mjefferson@gm.slc.edu | michael-jefferson-portfolio.vercel.app | github.com/jeffersonlegacy | axion-tax-beyond.vercel.app

Full-Stack TypeScript/JavaScript Engineer building **AI document workflows, local delivery commerce tools, and real-time WebRTC web applications** across **Next.js App Router, React 19, Vite, Node.js ESM, and Firebase**. Combines hands-on software delivery (2024–Present) with a background in psychology/neuroscience and 7 years of classroom mathematics instruction.

TECHNICAL STACK

AI, Document Workflows & Agentic Tooling: Gemini API (Pro / Flash / Vision), Vertex AI, OpenAI SDK & Codex, Document Triage & OCR Pipelines (Tesseract.js, ZXing, Vision LLMs), Ollama (Local LLM Inference), Multi-Agent Orchestration (OpenClaw), Structured JSON Schema Contracts, Context Routing (Agent Desk)

Frontend & Interactive UX: TypeScript, JavaScript (ES6+/ESM), Next.js (App Router, SSR/RSC), React 19, Vite, Tailwind CSS, Chart.js, Web Audio API, Remotion (Programmatic Video), Tldraw Canvas

Backend, Real-Time & Cloud: Node.js (ESM), Express 5, Firebase (Firestore, Realtime Database, Auth, Storage), LiveKit (WebRTC & RBAC JWTs), Twilio 10DLC SMS API & Webhooks, Resend, CloudConvert API, REST APIs

DevOps, QA & Infrastructure: Vercel, Render (render.yaml IaC), Docker, Playwright (Headless E2E & Visual QA), Vitest, Python (ReportLab PDF Synthesis), Sentry Observability, Debian Linux (zram tuning), Tailscale Mesh VPN, 2.5GbE LAN

SOFTWARE & APPLIED AI ENGINEERING EXPERIENCE

Full-Stack & Applied AI Product Engineer — Jefferson Legacy Systems (Independent Client & Product Engineering) | Vacaville, CA | 2024 – Present

- **Axion Tax & Beyond (axion-tax — Owner-Reviewed Client Prototype):** Built and deployed a full-stack tax intake portal (**Vite + React 19 SPA on Vercel** and **Node.js ESM Express 5 API**) featuring categorized upload lanes, client-side **SHA-256 file fingerprinting and filename-based document triage**, Schedule C & Write-Off Discovery wizards, and bilingual (EN/ES) Twilio SMS + Resend Email alerts.
- **Automated Visual QA & PDF Deliverable Pipeline:** Engineered a headless **Playwright** viewport capture workflow (output/playwright/) paired with a custom **Python PDF synthesis script (build_axion_pitch_deck.py)** to programmatically compile live application states into an executive pitch deck.
- **Tasty Cuts & More (tasty-cuts-and-more — Local Delivery Commerce & Owner Discovery Prototype):** Built a household freezer-pack delivery storefront and **48-question Web Audio voice discovery intake tool** in **React 19, Vite 8, and Firebase Hosting** with local promo validation (TASTY10 / FILLTHEFREEZER) and structured JSON/text brief export.
- **Tower Hour (tower-hour — Live Social Commerce & Multimodal Scanner Platform):** Developed a live-event shopping and product scanner application in **Next.js 16 App Router, TypeScript, Tailwind CSS, and Firebase** (Firestore + RTDB) across a 373-SKU catalog.
- **WebRTC RBAC & Twilio 10DLC SMS State Machine:** Implemented cryptographically signed **LiveKit RBAC JWTs** isolating broadcaster publish permissions from viewer subscribe tokens, alongside a bidirectional **Twilio 10DLC webhook pipeline** and **Sentry** observability.
- **Multimodal Product Scanner & Local Remotion Worker:** Combined **Tesseract.js, ZXing, and Vertex AI / Gemini** (/api/scan) to map photographed household products to catalog SKU replacements, and built a local **Remotion** worker for zero-cloud-cost video rendering.
- **Jefferson Super Reader 3000 (Document Vision & RSVP Instrument):** Built an asynchronous document conversion and reading pipeline integrating **CloudConvert** and **Gemini Vision APIs** with multi-word chunking, progressive WPM ramp, and a client-side **Optimal Recognition Point (ORP)** RSVP engine.
- **Mike Mac Mesh & Coach Mike AI:** Provisioned a headless **Debian Linux** inference server (tuned **zram** swap + **Ollama**) bridged via **2.5GbE LAN / Tailscale** to an M2 Mac mini with file-first context routing (agentdesk), alongside an interactive **Chart.js** role-alignment coaching visualizer.

Founder & STEM Systems Instructor — Tower Tutoring | Vacaville, CA | 2023 – Present

- Built an interactive collaborative tutoring web platform (**React, Vite, LiveKit WebRTC, Tldraw canvas, OpenAI SDK, Firebase**) and delivered 1-on-1 quantitative coaching, applying cognitive load principles to human-in-the-loop educational tools.

INSTRUCTIONAL & STAKEHOLDER COMMUNICATION EXPERIENCE

Mathematics Instructor & Education Specialist — Public School Districts (EOLA / VUSD / TUSD) | Solano & Alameda Counties, CA | 2018 – 2025

- Led 7 years of classroom Algebra and Geometry instruction and designed data-driven STEM curricula, translating complex quantitative frameworks into clear mental models for diverse students, parents, and administrative stakeholders.

EDUCATION

Sarah Lawrence College — Bronxville, NY | **Bachelor of Arts** (Psychology & Neuroscience Concentration; Kinetic Sculpture & Physical Computing)

LaGuardia Community College — Long Island City, NY | **Associate of Arts**